Corner-Adapted Quadrature for Rellich Normalization

Why the current rule fails at reentrant corners, and what replaces it

lappy development notes

Abstract

The Rellich identity computes $\|u\|_{L^2(\Omega)}$ from boundary Cauchy data alone. Its integrand carries a factor $r^{2\nu-2}$ at a corner of interior angle α, where $\nu = \pi/\alpha$; at a reentrant corner $\nu < 1$ and this is singular. We show (i) why the singularity can be removed *geometrically* by placing the Rellich reference point x_0 at the corner, (ii) why the current graded rule cannot integrate it when x_0 is placed elsewhere — the failure is truncation, not under-resolution — and (iii) that a corner-local Gauss–Jacobi rule in the variable $t = r^\sigma$ integrates it to machine precision with $O(10)$ nodes when σ rationalizes the corner's exponent family, and at a high-order algebraic rate otherwise — removing the constraint on x_0 in either case.

Read Section 7 before implementing. Sections 1–6 are the original analysis, and its central claim survived; three of its subsidiary conclusions did not. The substitution that rationalizes the exponent family is $t = r^{1/q}$ rather than $t = r^\nu$, and is unusable in that form for $q \geq 4$; the $2/\nu \in \mathbb{Z}$ condition is a limitation of $t = r^\nu$ specifically, not of the method; and the recipe of Section 6 omits a geometric cap on panel length whose absence costs twelve orders on some domains.

1 The identity and its corner integrand

For $-\Delta u = \lambda u$ in $\Omega \subset \mathbb{R}^2$ with $u = 0$ on $\partial\Omega$, the Rellich–Pohozaev identity with reference point $x_0 \in \mathbb{C}$ gives the L^2 norm as a pure boundary integral,

$$\|u\|^2_{L^2(\Omega)} \;=\; \frac{1}{2\lambda}\int_{\partial\Omega} \big((x - x_0)\cdot n\big)\Big(\frac{\partial u}{\partial n}\Big)^2 \,\mathrm{d}s \;=\; \frac{1}{2\lambda}\int_{\partial\Omega} r_N \Big(\frac{\partial u}{\partial n}\Big)^2 \,\mathrm{d}s, \tag{1}$$

with $r_N := (x - x_0)\cdot n$. The identity is exact for *every* x_0; the choice of x_0 affects only how hard the integral is to evaluate.

1.1 Local structure at a corner

Let the corner sit at the origin with interior angle α and edges along $\theta = 0$ and $\theta = \alpha$, and set

$$\nu \;:=\; \frac{\pi}{\alpha}, \qquad \nu_k := k\nu = \frac{k\pi}{\alpha}, \quad k = 1, 2, \ldots \tag{2}$$

The corner is *reentrant* when $\alpha > \pi$, i.e. $\nu < 1$; the 270° corner has $\nu = 2/3$ and the slit limit $\alpha \to 2\pi$ gives $\nu \to 1/2$. Any Dirichlet solution admits the local expansion

$$u(r,\theta) \;=\; \sum_{k\geq 1} c_k\, J_{\nu_k}\big(\sqrt{\lambda}\, r\big)\, \sin(\nu_k\theta). \tag{3}$$

On the edge $\theta = 0$ the outward normal is $n = -e_\theta$, so $u \equiv 0$ there forces the normal derivative to come entirely from the angular derivative:

$$\frac{\partial u}{\partial n}\bigg|_{\theta=0} = -\frac{1}{r}\frac{\partial u}{\partial \theta}\bigg|_{\theta=0} = -\sum_{k\ge 1} c_k\,\nu_k\,\frac{J_{\nu_k}(\sqrt{\lambda}r)}{r}. \tag{4}$$

Using $J_\mu(z) = (z/2)^\mu A_\mu(z^2)$ with A_μ entire and $A_\mu(0) \neq 0$,

$$\frac{\partial u}{\partial n}\bigg|_{\theta=0} = r^{\nu-1}F(r), \qquad F(r) = \sum_{k\ge 1} b_k\, r^{(k-1)\nu} A_{\nu_k}(\lambda r^2), \tag{5}$$

so that

$$\left(\frac{\partial u}{\partial n}\right)^2\bigg|_{\theta=0} = r^{2\nu-2}\,G(r), \qquad G(r) \;=\; F(r)^2 \;=\; \sum_{p\ge 0}\sum_{q\ge 0} g_{pq}\, r^{\,p\nu+2q}. \tag{6}$$

Two families of exponents appear in G: the *corner* exponents $r^{p\nu}$ generated by the expansion (3), and the *Bessel* exponents r^{2q} from the entire factors. This double structure is the crux of Section 3.1.

1.2 The geometric escape: $r_N \equiv 0$

Proposition 1. *If x_0 is placed at the corner, then $r_N \equiv 0$ on both straight edges emanating from it.*

Proof. On the edge $\theta = 0$, $x - x_0 = re^{i\cdot 0}$ is parallel to the edge, while $n = -e_\theta$ is perpendicular to it; hence $(x - x_0)\cdot n = 0$ pointwise, and likewise on $\theta = \alpha$. □

The singular factor $r^{2\nu-2}$ is therefore multiplied by an *exact* zero, and the corner contributes nothing: the boundary integral collapses onto the parts of $\partial\Omega$ where u is smooth. This is why placing x_0 at a reentrant corner yields machine precision with any reasonable quadrature. It is also why the escape does not generalize: x_0 is a single point, and a domain with several reentrant corners (H-shape, GWW) cannot zero r_N at all of them.

2 Why the graded rule fails for x_0 off the corner

For general x_0, r_N is a nonzero constant along each straight edge (it is the signed distance from x_0 to the edge's line), and the edge contributes

$$I_{\text{edge}} \;=\; \rho \int_0^R r^{2\nu-2} G(r)\,\mathrm{d}r, \qquad \rho := r_N = \text{const.} \tag{7}$$

This converges for $\nu > 1/2$, but the convergence is deceptive.

Lemma 1 (Mass concentration)**.** *The fraction of $\int_0^R r^{2\nu-2}\,\mathrm{d}r$ contributed by $r < \varepsilon$ is $(\varepsilon/R)^{2\nu-1}$.*

The exponent $2\nu - 1$ vanishes as $\nu \to 1/2$. Numerically, with $R = 1$ and $\varepsilon = 10^{-9}$:

α	ν	$2\nu-1$	mass below 10^{-9}
1.25π	0.800	0.600	4.0×10^{-6}
1.5π	0.667	0.333	1.0×10^{-3}
1.75π	0.571	0.143	5.2×10^{-2}
1.9π	0.526	0.053	0.336
1.99π	0.503	0.005	0.901

For a near-slit corner, *ninety percent of the integral lives within* 10^{-9} *of the corner.* lappy's Kress-graded rule clamps its innermost node at `quad._KRESS_TAU_FLOOR` $= 10^{-9}$ (a guard against a genuine defect: nodes built as $p_0 + \tau L\,\hat{d}$ in absolute coordinates round to exactly p_0 once τL falls below machine epsilon relative to $|p_0|$). The rule is thus *truncated*, not under-resolved: refining from 565 to 6790 nodes leaves the innermost node at 10^{-9} and the error unmoved. Observed errors track Lemma 1 over eight orders of magnitude, scaling linearly in $|x_0 - \text{apex}|$ with constant $\approx (10^{-9})^{2\nu-1}$.

Remark 1. The floor is a symptom of *coordinate representation*, not of floating-point resolution. The value $r = 10^{-30}$ is perfectly representable; it is $p_0 + \tau L\hat{d}$ that is not. A rule evaluated in corner-local (r,θ) never forms that difference. This distinction is what makes Section 3 possible.

3 A corner-local Gauss–Jacobi rule

Gauss–Jacobi quadrature with weight $(1-x)^a(1+x)^b$ integrates $\int_{-1}^{1}(1-x)^a(1+x)^b f(x)\,\mathrm{d}x$ exactly for f polynomial of degree $\le 2n-1$, and spectrally for f analytic. Mapping $r = R(1+x)/2$ and matching $b = \beta := 2\nu - 2$ to (7),

$$\int_0^R r^{2\nu-2}G(r)\,\mathrm{d}r \;=\; \left(\frac{R}{2}\right)^{2\nu-1}\sum_{i=1}^{n} w_i\,G(r_i), \qquad r_i = \tfrac{R}{2}(1+x_i). \tag{8}$$

Admissibility requires $\beta > -1$, i.e. $\nu > 1/2$: precisely the range of integrability, degenerating exactly at the slit.

Two properties matter as much as the accuracy:

1. **The nodes stay away from the corner.** The smallest Jacobi node sits at distance $O(n^{-2})$, not 10^{-9}: for $n = 16$ the innermost node is near 10^{-3}. The singularity is absorbed *analytically* into the weight rather than resolved by clustering, so neither the truncation floor nor the coordinate cancellation of Remark 1 ever arises.

2. **Only ν is needed.** $G(r) = (\partial_n u)^2 r^{2-2\nu}$ is formed numerically from black-box Cauchy data; no analytic knowledge of u is used. Since $\nu = \pi/\alpha$ comes exactly from the geometry, this is free — but it must be exact, see Section 4.

3.1 The multi-exponent obstruction and the substitution

Rule (8) is spectral only if G is analytic. By (6) it is not: G carries fractional powers $r^{p\nu}$. For a *single* mode ($c_k = 0$, $k \ge 2$) we have $G(r) = b_1^2 A_{\nu_1}(\lambda r^2)^2$, analytic in r^2, and (8) is spectral — but a single mode is not the situation of interest. For a general solution, convergence degrades to algebraic.

Proposition 2 (Rationalizing substitution)**.** *Under $t = r^\nu$, equivalently $r = t^{1/\nu}$, the edge integral becomes*

$$\int_0^R r^{2\nu-2} G(r)\,\mathrm{d}r \;=\; \frac{1}{\nu}\int_0^{R^\nu} t^{\,1-1/\nu}\,\widetilde{G}(t)\,\mathrm{d}t, \qquad \widetilde{G}(t) = \sum_{p,q} g_{pq}\,t^{\,p+2q/\nu}, \tag{9}$$

a Jacobi-weight integral with exponent $\beta' = 1 - 1/\nu$. The corner family $r^{p\nu}$ maps to the integer powers t^p.

Proof. $\mathrm{d}r = \frac{1}{\nu}t^{1/\nu-1}\,\mathrm{d}t$ and $r^{2\nu-2} = t^{2-2/\nu}$; the exponents add to $(2-2/\nu)+(1/\nu-1) = 1-1/\nu$. Substituting $r^{p\nu+2q} = t^{p+2q/\nu}$ into (6) gives $\widetilde{G}$. $\square$

Admissibility $\beta' > -1$ again requires exactly $\nu > 1/2$.

The caveat. The substitution rationalizes the corner family completely, but maps the Bessel family $r^{2q} \mapsto t^{2q/\nu}$, which is integral only when $2/\nu \in \mathbb{Z}$ — equivalently $\alpha = m\pi/2$ for an integer m. The 270° corner is such a case:

$$\alpha = \tfrac{3}{2}\pi \implies \nu = \tfrac{2}{3} \implies 2/\nu = 3 \in \mathbb{Z}, \qquad \widetilde{G}(t) = \sum g_{pq} t^{\,p+3q} \;\; \text{(a power series in } t\text{)}. \tag{10}$$

For generic α the residual exponents $2q/\nu$ are fractional, but they are *high order*: the leading one is $2/\nu > 2$, so $\widetilde{G} \in C^2$ at worst and Gauss convergence is algebraic of high order rather than spectral. This is confirmed in Section 5(c): the machine precision at $n = 8$ is specific to $\alpha = \frac{3}{2}\pi$, while generic α converges at roughly two digits per node doubling — still reaching 10^{-12} by $n = 32$, which is cheap enough that the distinction is of no practical consequence.

4 Sensitivity to ν

The weight exponent must use ν at full precision. Perturbing $\nu \to \nu + \delta$ leaves a residual factor $r^{-2\delta}$ in the integrand, which is unbounded as $r \to 0$ for $\delta > 0$ and destroys the analyticity the rule depends on. Measured at $\alpha = 1.5\pi$ (true $\nu = 2/3$), mode $(1, 1)$:

ν used in β	28 nodes	192 nodes
0.6667 (exact)	-7.5×10^{-15}	-3.1×10^{-15}
0.667	6.6×10^{-5}	1.7×10^{-5}
0.600	-1.8×10^{-2}	-4.7×10^{-3}

A relative error of 3×10^{-4} in ν costs four digits and reduces convergence to algebraic. Since $\nu = \pi/\alpha$ is exact from the geometry this costs nothing — but it rules out tabulated, rounded, or margin-padded exponents of the kind used for grading orders.

5 Numerical verification

All errors are relative, against closed-form sector eigenfunctions of unit norm (exact L^2 norms, so the reference is analytic rather than quadrature-based).

(a) Single mode, 270° corner, x_0 at the failing default. Cauchy data treated as a black box; $2n_{\text{edge}} + n_{\text{arc}}$ nodes total.

nodes	Gauss–Jacobi	existing Kress rule
16	-7.1×10^{-9}	—
28	-7.5×10^{-15}	—
2264	—	2.2×10^{-7} (plateau)

(b) Multi-exponent corner data. A five-term expansion (3) times an analytic tail, integrated against an mpmath reference at 40 digits, at $\nu = 2/3$:

n	Jacobi in r	Jacobi in $t = r^\nu$
8	-4.6×10^{-3}	4.0×10^{-15}
16	-1.2×10^{-3}	7.3×10^{-15}
64	-7.2×10^{-5}	1.6×10^{-13}

Plain Jacobi in r is algebraic, as Section 3.1 predicts; the substituted rule is spectral and *degrades* slowly past $n \approx 32$ from roundoff amplification, so small n is both cheaper and more accurate.

(c) Generic α: is $\nu = 2/3$ special? The same five-term data, substituted rule, against an *exact* reference (see the warning below).

α	$2/\nu$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
1.5π	3.000 (int.)	2.0×10^{-15}	2.0×10^{-15}	-2.6×10^{-14}	1.7×10^{-13}
1.33π	2.667	2.8×10^{-8}	2.7×10^{-10}	2.7×10^{-12}	5.1×10^{-14}
1.25π	2.500	4.7×10^{-8}	5.1×10^{-10}	5.9×10^{-12}	5.0×10^{-14}
1.6π	3.200	-5.6×10^{-9}	-3.4×10^{-11}	-1.6×10^{-13}	-7.7×10^{-13}
1.7π	3.400	-5.2×10^{-9}	-2.7×10^{-11}	-3.4×10^{-13}	-9.8×10^{-13}
1.85π	3.700	-1.5×10^{-9}	-6.0×10^{-12}	1.9×10^{-14}	-2.5×10^{-12}
1.95π	3.900	-1.4×10^{-10}	-2.0×10^{-13}	-4.0×10^{-13}	-1.2×10^{-11}

Only $\alpha = \frac{3}{2}\pi$ is machine-exact at $n = 8$, confirming Section 3.1: the integer $2/\nu$ is genuinely special. Every other angle converges at $\approx$ two digits per doubling — high-order algebraic, exactly as predicted — and all reach 10^{-12} or better by $n = 32$. The mild degradation past $n \approx 32$ is roundoff, not truncation.

Warning: mpmath's `quad` is not a safe reference here. The endpoint behaviour $r^{2\nu-2} \to r^{-1}$ as $\nu \to \frac{1}{2}$ defeats `mp.quad` even at 40 digits. Measured against the exact termwise value $\int_0^1 r^a \, \mathrm{d}r = (a+1)^{-1}$, its error reaches -4.7×10^{-5} at $\alpha = 1.85\pi$ and -4.3×10^{-2} at $\alpha = 1.95\pi$. Using it as ground truth produces convincing but entirely spurious "plateaus" in the rule under test. Any reference for this integral should be closed-form.

A trap worth recording. At $\alpha = 3\pi/2$ the two edges carry $r_N = \operatorname{Im} x_0$ and $-\operatorname{Re} x_0$ against identical $(\partial_n u)^2$, so any x_0 on the diagonal $\operatorname{Re} x_0 = \operatorname{Im} x_0$ cancels the singular contribution *identically.* A test using such an x_0 silently measures the smooth part of the boundary only. Corner-quadrature experiments should always report the edge/arc split.

6 Recommended construction

For each corner with $\nu < 1$, and each straight edge meeting it:

1. Work in corner-local coordinates (r, θ); never form $p_0 + \tau L\hat{d}$.
2. Panel the edge, substitute $t = r^\nu$ with $\nu = \pi/\alpha$ exact.
3. Apply Gauss–Jacobi with $\beta' = 1 - 1/\nu$ and $n = 8$–16 nodes.

Smooth portions of $\partial\Omega$ retain plain Gauss–Legendre. Edges meeting two singular corners are split, one panel per endpoint.

This removes the requirement that x_0 sit at a corner, and with it the obstruction to multi-corner domains: no partition of unity over corners is needed. The requirement $\nu > 1/2$ is not restrictive — it is exactly the condition for (7) to converge at all — but the slit limit $\nu \to 1/2$ remains delicate, both weights degenerating as $\beta, \beta' \to -1$.

Not yet verified. A real `FourierBesselBasis` through this rule (as opposed to closed-form or synthetic data), and a genuinely multi-corner domain.

7 Addendum: what implementation changed

The analysis above was written before the rule was built. Its core result held: a corner-adapted rule reaches 10^{-14} where the graded rule plateaus at 10^{-7}, with fewer nodes. What follows corrects the parts that did not survive contact with a real domain, and records what could only be learned by measuring. Everything here is implemented in `lappy.quad` and `lappy.eigfun_integrals`, and tested in `tests/test_eigfun_integrals.py`.

7.1 The substitution is $t = r^{1/q}$, and $t = r^\nu$ is a trap for curved edges

Section 3.1 identified the exponent family as $\{k\nu + 2q\}$ and concluded that $t = r^\nu$ rationalizes it when $2/\nu \in \mathbb{Z}$ — for reentrant α, only $\alpha = 3\pi/2$. That is correct *for a straight edge*, and it is the wrong family for a curved one.

On a curved edge two things change, both adding *integer* powers of arclength s. First, r_N is no longer constant: with x_0 at the corner it is $(\kappa_0/2)s^2 + O(s^3)$ — vanishing to second order rather than identically, so Proposition 1's escape degrades gracefully rather than failing. Second, $r = |x(s) - \text{corner}| = s - (\kappa_0^2/24)s^3 + \cdots$, so r is not arclength. The family becomes $\{k\nu + m\}$, $m \in \mathbb{Z}_{\geq 0}$, including *odd* powers. Under $t = r^\nu$ that maps to $\{k + m/\nu\}$ whose leading residual $t^{1/\nu}$ has $1/\nu \in (1, 2)$: only C^1. Measured, it never reaches 10^{-13} at any order up to 64. Under $t = r^{1/q}$ with $\nu = p/q$ in lowest terms it maps to $\{kp + mq\}$ — all integers — and is exact by order 6–14.

But $t = r^{1/q}$ is unusable as written. Since $\tau = t^q$, the innermost node behaves as $\tau_{\min} \sim t^q_{\min}$:

α	q	$\tau_{\min}$ at order 24	
$3\pi/2$	3	1.4×10^{-8}	
$5\pi/4$	5	1.1×10^{-10}	(below the floor)
$7\pi/4$	7	4.7×10^{-19}	
$11\pi/6$	11	1.6×10^{-29}	

Those nodes round onto the corner once mapped through a segment's parametrization — the very defect Remark 1 identifies. The rule is exact as an abstract statement about $[0, 1]$ and worthless on a boundary.

7.2 What works: exactness from the weights, node placement from ν

The resolution separates the two roles the substitution was doing. Nodes are placed by $t = r^{\nu}$, whose crowding is mild ($\tau_{\min}$ stays at 10^{-5}–10^{-7}). Exactness comes from the *weights*: the exponent set $\{\gamma + j\nu + m\}$ is known, and every moment is closed-form, $\int_0^1 t^e \, dt = (e + 1)^{-1}$, so the weights are obtained by solving a linear system. This also covers irrational ν, which the substitution cannot touch at all — and irrational ν is the *generic* case for a curved corner, whose angle is fixed by the geometry rather than chosen (two overlapping circles give $\nu = 0.657360$).

One trap: solving the square system (as many exponents as nodes) is exact on its span but has cond $\sim 10^{13}$ at order 12 and 10^{19} at order 16, with weights reaching -10^4. Since $\sum |w|$ multiplies every roundoff error in the integrand, that imports a 10^{-12} floor. Using *fewer exponents than nodes* — a minimum-norm least-squares solve — keeps $\sum |w| = 1.0$ with exactness intact.

7.3 The edge type matters, and conflating them costs seven orders

A straight edge admits only *even* integer powers (r_N exactly constant, r exactly arclength), so $\{\gamma + j\nu + 2q\}$; odd powers are curved-only. Scored against the curved set, the substitution rule at $\alpha = 3\pi/2$ reads 10^{-8} — against a measured 2.4×10^{-15} on a real sector. Any error indicator must use the family belonging to the edge it is scoring.

7.4 Panel length must be capped by clearance, not by a fraction

Section 6's recipe panels the edge but says nothing about how long a panel may be. The corner expansion is valid only within the largest disk about the corner inside Ω; a panel reaching past it is under-resolved, because the rule clusters its nodes *at* the corner and cannot resolve the $\sqrt{\lambda}$ oscillation over the remaining arclength. Measured on a $1 \times N$ polyomino strip with one cell below an end (so the corner's edge has length $N - 1$ against a clearance of 1), against its exact eigenfunction:

N	no cap	$0.9\times$ clearance
2	6.5×10^{-14}	5.3×10^{-15}
8	2.3×10^{-2}	1.6×10^{-15}
16	7.7×10^{-2}	1.3×10^{-15}
24	7.4×10^{-3}	4.4×10^{-16}

Twelve orders. A fixed *fraction* of the edge cannot substitute — being a fraction of the edge, it grows with the edge and stays too long (0.25 of the edge still gives 2.6×10^{-4} at $N = 16$). The cap must be geometric.

7.5 Two properties that look like bugs and are not

$\sum_i w_i = 1$ holds only when the constant function lies in the rule's exact class. Renormalizing would destroy exactness on the singular class the rule exists for; $\sum_i w_i - 1$ is instead its cheapest self-diagnostic. And accuracy is *not monotone* in order: past a ν-dependent threshold the integrand's dynamic range ($\tau^{2\nu-2} \sim 6 \times 10^8$ at $\tau \sim 10^{-9}$ against a correspondingly tiny weight) makes roundoff dominate. At $\nu = 0.526$ the best order is 16 while the floor-based cap is 75, and using the cap gives 4.2×10^{-9}. Order selection must therefore scan a calibrated curve, not increase the order until a target is met.

7.6 Scope, sharpened

The rule is matched to *eigenfunction*-level Cauchy data, whose local structure is purely the corner family. It is *not* appropriate for a basis-level $N \times N$ Gram matrix: columns centred at other corners are plain analytic at this one, with $O(1)$ amplitude at every corner simultaneously. Measured, the corner rule beats the graded rule by 3–8 orders on corner-family blocks and loses by 2–4 on mixed ones. That is why `lappy` retired the basis-level path rather than adapting it.

Finally, a warning about instruments. `mpmath.quad` is not a valid reference for these integrands: as $\nu \to 1/2$ the endpoint behaviour approaches r^{-1} and it errs by 4×10^{-2} even at 40 digits, manufacturing convincing but entirely spurious convergence plateaus. Every reference used in the tests is closed-form.